Project Title: Weather Long-Term Time Series Forecasting Using Machine Learning and Deep Learning Models

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1. Overview

The project aims to develop a weather forecasting system using machine learning and deep learning models (LSTM, GRU, MLP). By gathering high-frequency meteorological data, our objective is to analyze long-term weather patterns and generate accurate forecasts for the next 24 hours. At this stage, approximately 60% of the work has been completed. This includes data preprocessing, model development, and initial performance evaluation. The remaining tasks involve refining the models, and completing all documentation and paper writing before the April 8, 2025 deadline

2. Work Completed

- **Data Preprocessing and Exploratory Analysis**
  - Cleaned and structured a large, high-frequency weather dataset.
  - Applied MinMax scaling to normalize data.
  - Prepared sequence-based inputs for time-series forecasting.
- **Model Development and Training**

  Implemented three models:
  - LSTM: Bidirectional architecture with Dropout, BatchNormalization, and Adam optimization.
  - GRU: Deep recurrent model with stacked layers and tuned hyperparameters.
  - MLP: A fully connected network as a baseline.
  - Trained each model for 100 epochs, using early stopping to prevent overfitting.
  - Employed a learning rate reduction strategy (ReduceLROnPlateau).
- **Initial Model Evaluation**
  - Assessed performance via MSE and MAE.
  - Compared outcomes to identify areas needing further optimization.
  - Achieved promising initial results, but additional fine-tuning is required.

3. Challenges Encountered

- **High Computational Requirements**
  - Training deep networks on time-series data demanded significant GPU usage.
  - Resolved by adjusting batch sizes, modifying network architectures, and using early stopping.
- **Optimizing Sequence Length and Forecasting Window**
  - Determining the right time-step for reliable long-term predictions proved challenging.
  - Further experiments are needed to balance model complexity and accuracy.
- **Model Performance Variability**
  - LSTM and GRU performance varied with different hyperparameter configurations.
  - Additional tuning is necessary to achieve consistent results on unseen data.

4. Next Steps (March 6 – April 8)

- **Fine-Tuning and Hyperparameter Optimization**
  - Continue experimenting with sequence lengths, dropout rates, and layer configurations.
  - Investigate attention mechanisms to enhance long-term forecasting accuracy.
- **Paper Writing and Report Finalization**
  - Develop and refine the project paper, emphasizing key findings and methodologies.
  - Prepare thorough documentation for all project components.

5. Conclusion

This project has reached the halfway mark, with data preparation, model construction, and preliminary training completed. The remaining tasks involve further optimization by choosing the best model that suits our data and the development of a comprehensive paper. Progress is on schedule, and the project is on track for completion by April 8, 2025.